

Numbering System Conversions

Show Work! Point deductions taken if work not shown

1. Convert 19_{10} to binary

(128x0) (64x0) (32x0) (16x1) (8x0) (4x0) (2x1) (1x1)

$$16 + 2 + 1 = 19$$

$$19^{10} = 00010011^2$$

2. Convert 158_{10} to binary

(128x1) (64x0) (32x0) (16x1) (8x1) (4x1) (2x1) (1x0)

$$128 + 16 + 8 + 4 + 2 = 158$$

$$158^{10} = 10011110^2$$

3. Convert 121_{10} to binary

(128x0) (64x1) (32x1) (16x1) (8x1) (4x0) (2x0) (1x1)

$$64 + 32 + 16 + 8 + 1 = 121$$

$$121^{10} = 01111001^2$$

4. Convert 31_{10} to binary

(128x0) (64x0) (32x0) (16x1) (8x1) (4x1) (2x1) (1x1)

$$16 + 8 + 4 + 2 + 1 = 31$$

$$31^{10} = 00011111^2$$

5. Convert 69_{10} to binary

(128x0) (64x1) (32x0) (16x0) (8x0) (4x1) (2x0) (1x1)

$$64 + 4 + 1 = 69$$

$$69^{10} = 01000101^2$$

6. Convert 164_{10} to binary

(128x1) (64x0) (32x1) (16x0) (8x0) (4x1) (2x0) (1x0)

$$128 + 32 + 4 = 164$$

$$164^{10} = 10100100^2$$

7. Convert 91_{10} to hexadecimal

$$91/16 = 5 \text{ R } 11. \text{ 11} = \text{B}, 5 = 5$$

$$5\text{B}^{16}. (5 \times 16) + (11 \times 1) 80 + 11 = 91$$

$$91^{10} = 5\text{B}^{16}$$

8. Convert 98_{10} to hexadecimal

$$98/16 = 6 \text{ R } 2. \text{ 6}, 2$$

$$62^{16}. (6 \times 16) + (2 \times 1) 96 + 2 = 98$$

$$98^{10} = 62^{16}$$

9. Convert 69_{10} to hexadecimal

$$69/16 = 4 \text{ R } 5. \text{ 4}, 5$$

$$45^{16}. (4 \times 16) + (5 \times 1) 64 + 5 = 69$$

$$69^{10} = 45^{16}$$

10. Convert 125_{10} to hexadecimal

$$125/16 = 7 \text{ R } 13. \text{ 7}, 13 = \text{D}$$

$$7\text{D}^{16}. (7 \times 16) + (13 \times 1) 112 + 13 = 125$$

$$125^{10} = 7\text{D}^{16}$$

11. Convert 204_{10} to hexadecimal

$$204/16 = 12 \text{ R } 12. \text{ 12} = \text{C}$$

$$\text{CC}^{16}. (12 \times 16) + (12 \times 1) 192 + 12 = 204$$

$$204^{10} = \text{CC}^{16}$$

12. Convert 12_{10} to hexadecimal

$$12/16 = 0 \text{ R } 12. \text{ 12} = \text{C}$$

$$\text{C}^{16}. (0 \times 16) + (12 \times 1) 12 = 12$$

$$12^{10} = \text{C}^{16}$$

13. Convert 1111_2 to hexadecimal

$$1111^2 = 8 + 4 + 2 + 1 = 15. 15 = F$$

$$1111^2 = F^{16}$$

14. Convert 1011_2 to hexadecimal

$$1011^2 = 8 + 2 + 1 = 11. 11 = B$$

$$1011^2 = B^{16}$$

15. Convert 11010101_2 to hexadecimal

$$1101 = 8 + 4 + 0 + 1 = 13 = D$$

$$0101 = 0 + 4 + 0 + 1 = 5$$

$$11010101^2 = D5^{16}$$

16. Convert 11011011_2 to hexadecimal

$$91/16 = 5 \text{ R } 11. 11 = B, 5 = 5$$

$$5B^{16}. (5 \times 16) + (11 \times 1) 80 + 11 = 91$$

$$91^{10} = 5B^{16}$$

17. Convert $3D_{16}$ to binary

$$3^{16} = 3^{10}. 3 = 0011^2$$

$$D = 13. 13^{10} = 1101^2$$

$$3D^{16} = 00111101^2$$

18. Convert $1A_{16}$ to binary

$$1^{16} = 1^{10}. 1 = 0001^2. A = 10$$

$$10^{10} = 1010^2$$

$$1A^{16} = 00011010^2$$

19. Code the following word in hexadecimal then to binary. Be sure to watch video before completing this conversion

E G G

E = 45¹⁶. 4 = 0100, 5 = 0101

45¹⁶ = 01000101²

G = 47¹⁶. 6 = 0100, 7 = 0111

47¹⁶ = 01000111²

E = 45, G = 47, G = 47

454747¹⁶ = 010001010100011101000111²